Algebra 1 – Module 11 Practice Test •
WORKED KEY

Full Name: ______________________________
Period: _________

This practice is built exactly like the real test — same questions, same order, different numbers. Work every problem, then check the worked key. Anything you miss is exactly what to study.

1. For the quadratic $y = x^2 + 10x + 9$, find the axis of symmetry, the vertex, and the y-intercept.

Answer: axis $x = -5$; vertex $(-5, -16)$; y-intercept $(0, 9)$

2. Solve by factoring: $x^2 - 5x - 14 = 0$

Answer: $(x - 7)(x + 2) = 0 \rightarrow x = 7$ or $x = -2$

3. Solve by factoring: $3x^2 + 5x - 2 = 0$

Answer: $(3x - 1)(x + 2) = 0 \rightarrow x = 1/3$ or $x = -2$

4. Graph the function. Label the vertex and the axis of symmetry: $y = -x^2 + 4x - 3$

Answer: axis $x = 2$; vertex $(2, 1)$; y-int $(0, -3)$; zeros $x = 1, 3$ — opens DOWN since $a = -1$

5. Solve using square roots: $5x^2 - 245 = 0$

Answer: $x^2 = 49 \rightarrow x = \pm 7$

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6. Solve by completing the square: $x^2 - 10x - 11 = 0$

Answer: $x^2 - 10x + 25 = 36 \rightarrow (x - 5)^2 = 36 \rightarrow x - 5 = \pm 6 \rightarrow x = 11$ or $x = -1$

7. Solve using the quadratic formula: $3x^2 - 7x - 6 = 0$

Answer: $x = (7 \pm \sqrt{121})/6 = (7 \pm 11)/6 \rightarrow x = 3$ or $x = -2/3$

8. Find the discriminant, then state how many real solutions the equation has: $2x^2 - 3x + 5 = 0$

Answer: $b^2 - 4ac = 9 - 40 = -31 < 0 \rightarrow$ NO real solutions

9. Find the zeros and the axis of symmetry: $y = (x - 3)(x + 9)$

Answer: zeros $x = 3, -9$; axis halfway: $x = -3$

10. Error Analysis — Gus solved $x^2 = 144$ and wrote $x = 12$. Identify his mistake, then give the full solution.

Answer: he forgot the negative root — square roots give TWO answers: $x = \pm 12$

11. Solve using ANY method you choose: $x^2 - 15x + 56 = 0$

Answer: factors nicely: $(x - 7)(x - 8) = 0 \rightarrow x = 7$ or $x = 8$ (any valid method accepted)

12. A model rocket is launched from a 5-foot platform, so its height in feet after t seconds is $h = -16t^2 + 80t + 5$. Find the time when the rocket reaches its maximum height, and find that maximum height.

Answer: $t = -80/(-32) = 2.5$ seconds; $h = -16(6.25) + 80(2.5) + 5 = -100 + 200 + 5 = 105$ feet

13. A rectangular garden is 7 feet longer than it is wide. Its area is 60 square feet. Write and solve a quadratic equation to find the garden's dimensions.

Answer: $w(w + 7) = 60 \rightarrow w^2 + 7w - 60 = 0 \rightarrow (w + 12)(w - 5) = 0 \rightarrow w = 5$ (reject -12). Dimensions: 5 ft by 12 ft

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Cumulative Review — ACT Style. Circle the best answer.

14. Which is the COMPLETE factorization of $4x^2 + 12x + 8$?

Ⓐ $(4x + 4)(x + 2)$
Ⓑ $2(2x + 2)(x + 2)$
Ⓒ **$4(x + 1)(x + 2)$**
Ⓓ $(x + 2)(x + 4)$

Answer: GCF of 4 must come out first, and each factor must be fully factored

15. The population of a colony is modeled by $y = 500(1.45)^x$. What is the percent INCREASE each year?

Ⓔ 145%
Ⓕ **45%**
Ⓖ 1.45%
Ⓗ 55%

Answer: growth factor $1.45 = 1 + 0.45$

16. Simplify: $10/\sqrt{5}$

Ⓐ $10\sqrt{5}$
Ⓑ $\sqrt{5}/10$
Ⓒ **$2\sqrt{5}$**
Ⓓ $5\sqrt{5}$

Answer: multiply by $\sqrt{5}/\sqrt{5}$: $10\sqrt{5}/5 = 2\sqrt{5}$